MULT20015 Elements to Quantum Computing

Introduction to the Quantum User Interface (QUI)

The QUI is a web-based graphical user interface to program, simulate and analyse quantum circuits. It was developed by the Hollenberg group at the University of Melbourne. The QUI allows the users to specify qubit number, build and simulate quantum circuits by easy drag-and-drop placement of quantum gates, and examine the quantum state at every time step in the circuit/program. The latter feature is critical to understanding QC, and distinguishes the QUI from other on-line QC programing/simulation tools.

Sign up and start the QUI

The QUI is accessed through a web-based interface (quispace.org – click on blue QUI logo). As per the subject announcement you should have signed up prior to the Lab-1 class – if you haven't please let a demonstrator know and follow these steps.

Step 1: Open a web browser (preferably Google Chrome or Firefox), go to quispace.org.
Step 2: You will need to create an account to access QUI for the first time.

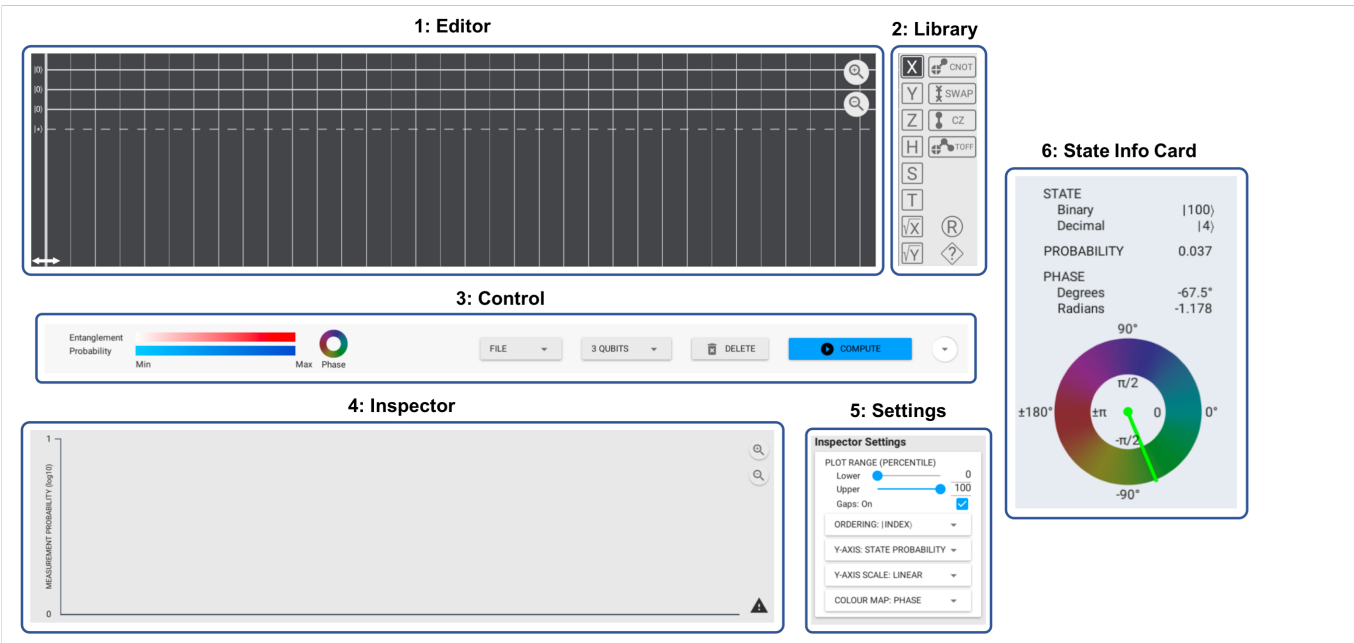
Note: In order to access expanded capabilities of the QUI you must use your University of Melbourne email address as your login name.

Follow the steps to create your account (including answering a few simple questions about use and education level).

Step 3: Once you have signed-up, start the QUI.

The QUI panels

Shown below are the various QUI panels.



Editor (Panel 1): This is where a quantum circuit is programed by dragging and dropping from the gate Library (Panel 2). On the left side, the initial states of three qubits (for this example) are shown in the default |0⟩ state. You can add more qubits using the |*> icon (or QUBITS in the Control panel). The vertical lines separate time steps left to right. The thicker vertical with the left-right arrow is the "time-slider", which can be moved to analyse the quantum state at different times in the quantum circuit. The time-slider will also display the degree of entanglement at various parts of the quantum circuit (Lab 2).

Library (Panel 2): This panel shows the gate library (quantum operations), including measurement. Hovering the cursor over any single-qubit gate will display the animation of the corresponding operation on the Bloch sphere. Select one of these gates and place it in the Editor panel at the desired location(s). There is even a paint feature.

Control (Panel 3): The colour scales on the left side indicate the min-max range of various output parameters (probability, entanglement, phase). Next is a drop-down FILE menu (New, Save, Load, Save As). QUBITS allows the users to select the number of qubits. DELETE removes selected quantum gates from the circuit. COMPUTE will run the current circuit and display the results in the Inspector panel (more accurately, the circuit is sent to our quantum simulator and the entire set of results (all time points) is sent back to your QUI session for analysing – please don't sit there and hit compute a million times).

Inspector (Panel 4): Here you can visualise the quantum state (after compute) at any time point corresponding to the position of the sliding time-slider. By default, the horizontal axis is ordered according to the state index in the computational basis (|11> = |3> etc), and the vertical axis by default gives the probability of each state. Detailed information about the state amplitudes (magnitude and phase) is given in the State Info Card that pops up when the mouse hovers over a state in the histogram. Note the hazard warning symbol bottom right – this indicates the current Inspector plot does not contain up-to-date data (i.e. with respect to changes in the circuit shown in the Editor).

Settings (Panel 5): Here the user has additional control over the various plotting options in the Inspector, such as PLOT RANGE, ORDERING, and AXIS (data) options.

State Info Card (Panel 6): This panel appears when the cursor is on a particular state in the Inspector panel and will display all information about the state's identity (binary and integer index), complex valued amplitude (magnitude and phase), and probability.